

Key points:

- The ***trp* operon**, found in *E. coli* bacteria, is a group of genes that encode biosynthetic enzymes for the amino acid tryptophan.
- The *trp* operon is expressed (turned "on") when tryptophan levels are low and repressed (turned "off") when they are high.
- The *trp* operon is regulated by the ***trp* repressor**. When bound to tryptophan, the *trp* repressor blocks expression of the operon.
- Tryptophan biosynthesis is also regulated by **attenuation** (a mechanism based on coupling of transcription and translation).

What is the *trp* operon?

Bacteria such as *Escherichia coli* (a friendly inhabitant of our gut) need amino acids to survive—because, like us, they need to build proteins. One of the amino acids they need is tryptophan.

If tryptophan is available in the environment, *E. coli* will take it up and use it to build proteins. However, *E. coli* can also make their own tryptophan using enzymes that are encoded by five genes. These five genes are located next to each other in what is called the ***trp* operon**.

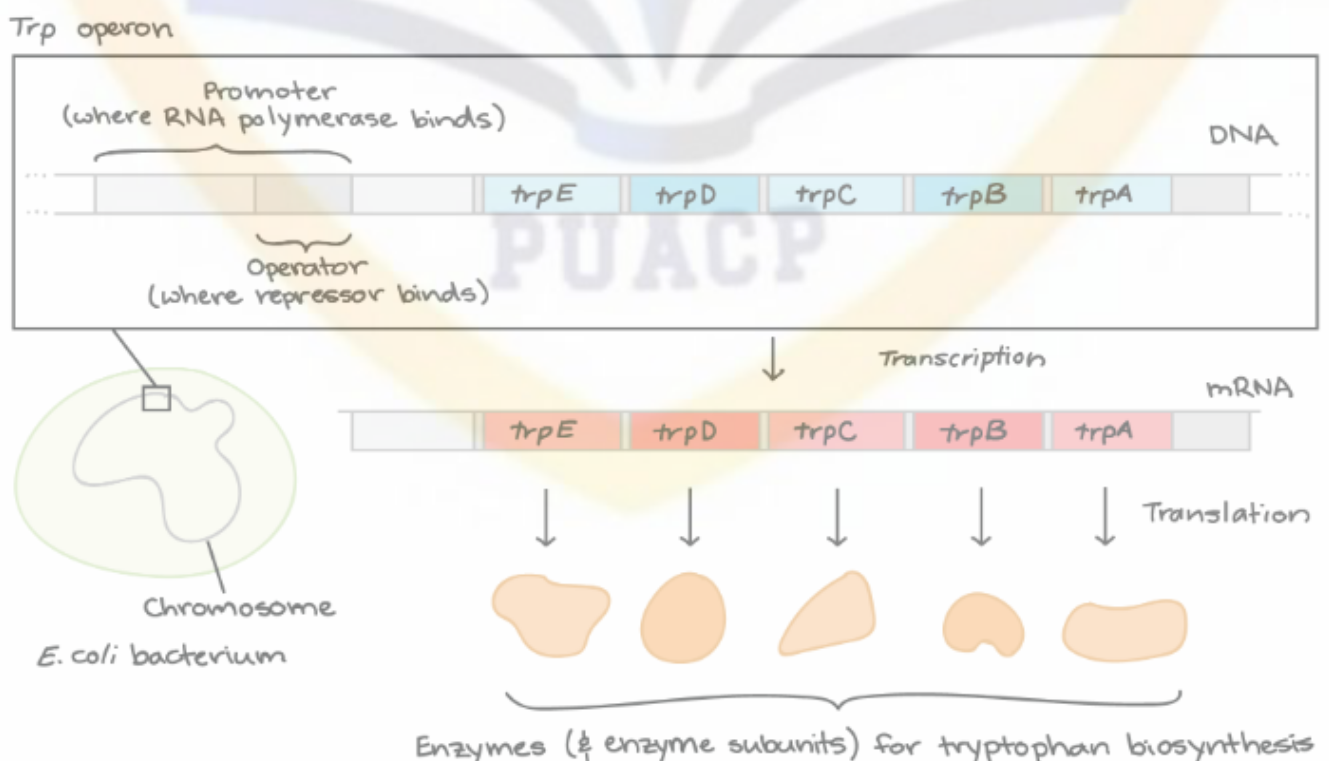
[\[What's an operon?\]](#)

If tryptophan is present in the environment, then *E. coli* bacteria don't need to synthesize it, so transcription of the genes in the *trp* operon is switched "off." When tryptophan availability is low, on the other hand, the operon is switched "on," the genes are transcribed, biosynthetic

enzymes are made, and more tryptophan is produced.

Structure of the *trp* operon

The *trp* operon includes five genes that encode enzymes needed for tryptophan biosynthesis, along with a **promoter** (RNA polymerase binding site) and an **operator** (binding site for a repressor protein). The genes of the *trp* operon are transcribed as a single mRNA.

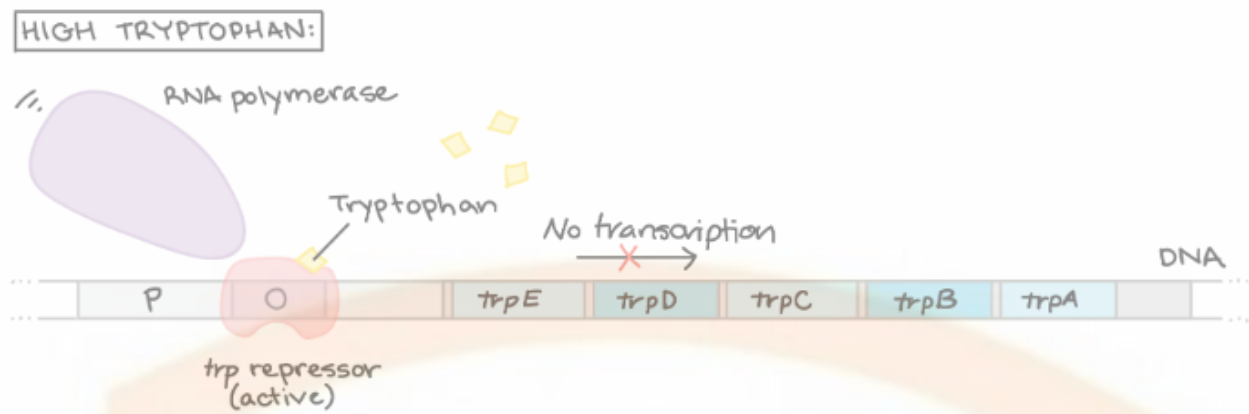


Turning the operon "on" and "off"

What does the operator do? This stretch of DNA is recognized by a regulatory protein known as the ***trp* repressor**. When the repressor binds to the DNA of the operator, it keeps the operon from being transcribed by physically getting in the way of RNA polymerase, the transcription enzyme.

*[Where does the *trp* repressor come from?]*

The *trp* repressor does not always bind to DNA. Instead, it binds and blocks transcription only when tryptophan is present. When tryptophan is around, it attaches to the repressor molecules and changes their shape so they become active. A small molecule like tryptophan, which switches a repressor into its active state, is called a **corepressor**.



When there is little tryptophan in the cell, on the other hand, the *trp* repressor is inactive (because no tryptophan is available to bind to and activate it). It does not attach to the DNA or block transcription, and this allows the *trp* operon to be transcribed by RNA polymerase.



In this system, the *trp* repressor acts as both a sensor and a switch. It senses whether tryptophan is already present at high levels, and if so, it switches the operon to the "off"

position, preventing unnecessary biosynthetic enzymes from being made.

More *trp* operon regulation: Attenuation

Depending on the class you're taking, or on your own interests, you may also have heard about another form of *trp* operon regulation called **attenuation**.

Like regulation by the *trp* repressor, attenuation is a mechanism for reducing expression of the *trp* operon when levels of tryptophan are high. However, rather than blocking *initiation* of transcription, attenuation prevents *completion* of transcription.

When levels of tryptophan are high, attenuation causes RNA polymerase to stop prematurely when it's transcribing the *trp* operon. Only a short, stubby mRNA is made, one that does not encode any

tryptophan biosynthesis enzymes. Attenuation works through a mechanism that depends on coupling (the translation of an mRNA that is still in the process of being transcribed). [\[See the details of attenuation\]](#)

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